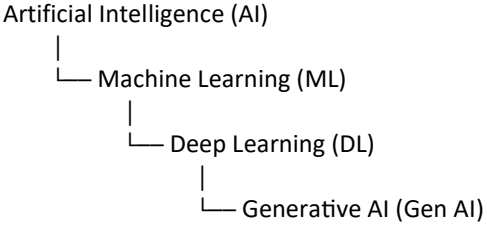
Definitions:

Definitions of: Artificial Intelligence, Machine Learning, Deep Learning & Generative AI

Hierarchy:



Each one is a **subset** of the one before it — narrowing in scope but increasing in capability for specific tasks (especially content creation, in Gen AI's case).

Layer	Definition	Example 1	Example 2
AI	Learning to **Simulate** human intelligence	Google Maps	Siri/Alexa
ML	Learning to **Predict/Classify** using data patterns	Spam filters	Netflix recommendations
DL	Learning to **Extract features automatically** using neural networks	Face unlock	Google Translate
Gen AI	Learning to **Create/Generate** new content	ChatGPT/Claude	DALL·E

1. Artificial Intelligence (AI)

The broad field of building machines/systems that can perform tasks requiring human-like intelligence — such as reasoning, learning, understanding language, and decision-making.

AI is the umbrella concept — machines simulating human intelligence.

Examples:

- **Google Maps** – uses AI for route optimization and traffic prediction
- **Siri/Alexa** – AI-powered virtual assistants that understand and respond to voice commands

2. Machine Learning (ML)

A subset of AI where systems learn patterns from data and improve their performance over time — without being explicitly programmed for every rule.

ML is how machines learn from experience (data) instead of fixed instructions.

Examples:

- **Spam filters (Gmail)** – learns from labeled emails to classify spam vs. not spam
- **Netflix/Amazon recommendations** – learns user behavior patterns to suggest movies/products

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3. Deep Learning (DL)

A subset of Machine Learning that uses multi-layered artificial neural networks to automatically learn complex patterns from large amounts of data.

DL is ML using deep neural networks — inspired by how the human brain processes information.

Examples:

- **Face unlock on smartphones** – uses CNNs (Convolutional Neural Networks) to recognize facial features
- **Google Translate** – uses deep neural networks (RNNs/Transformers) for language translation

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4. Generative AI (Gen AI)

A subset of Deep Learning focused on creating new content — text, images, audio, code, video — by learning patterns from existing data.

Gen AI doesn't just analyze or predict — it creates new, original outputs.

Examples:

- **ChatGPT/Claude** – generates human-like text responses, essays, code
- **DALL·E / Midjourney** – generates original images from text prompts

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